Level 2 Functional Skills Mathematics
Sample 3

City & Guilds
A City & Guilds Group Business

Version 1-1

Duration: 25 minutes
Total marks: 15 marks

SECTION 1 – CALCULATOR NOT PERMITTED

MODEL SOLUTION SCORE
15 / 15 marks (100% Pass)
25 minutes · Verified by MFS

Candidate name (first, last)

First: MASTER

Last: FUNCTIONAL SKILLS

Candidate enrolment number: MFS4748

Date of birth (DDMMYYYY): 01012000

Assessment date (DDMMYYYY): 20102026

Centre number: MFS0001

Candidate signature and declaration*: Master Functional Skills

***I declare that I had no prior knowledge of the questions in this assessment and that I will not share information about the questions.**

Please check that your name is correctly printed on the candidate barcode label. If not, please tell the invigilator before the start of the exam.

You should have the following for this assessment:

- a pen with black or blue ink
- a pencil
- an eraser
- a 30cm ruler.



You must NOT use a protractor.
You must NOT use a calculator for Section 1.

General instructions

- Read through each question carefully.
- You may use a dictionary.
- Write all your answers in this booklet.
- Check your calculations and check that your answers make sense.

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SECTION 1 – CALCULATOR NOT PERMITTED

There are **15** marks available in this section.

You should check all your work as you go along.

You must **not** use a calculator in this section.



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Q1

531 x 1.4 =

Method: 531 × 14 = (531 × 10) + (531 × 4) = 5310 + 2124 = 7434

Adjust for 1 decimal place: 7434 ÷ 10 = 743.4

[BI]

743.4

_____

(1 mark)

Q2

5^4 =

Method: 5^4 = 5 × 5 × 5 × 5 = 25 × 25 = 625

[BI]

625

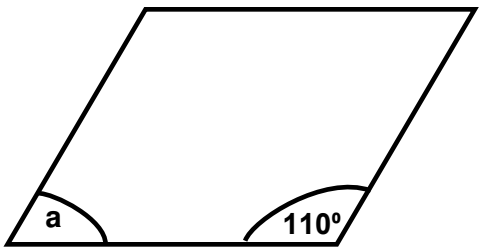
_____

(1 mark)

Q3

The following diagram shows a parallelogram.

Diagram not to scale



What is the size of angle **a?**

[BI]

Co-interior angles sum to 180°: a = 180° - 110° = 70°

Angle **a** = 70 °

(1 mark)

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Q4

Which one of the following works out to the largest number?

(tick one box)

A $\frac{3}{4}$ x 8 = = 6 ☐

B $\frac{2}{6}$ x 12 = = 4 ☐

C $\frac{3}{2}$ x 10 = = 15 ☑ 15 is largest --> Option C [B1]

D $\frac{3}{12}$ x 36 = = 9 ☐

(1 mark)

Q5

What is 52% as a fraction in its lowest terms?

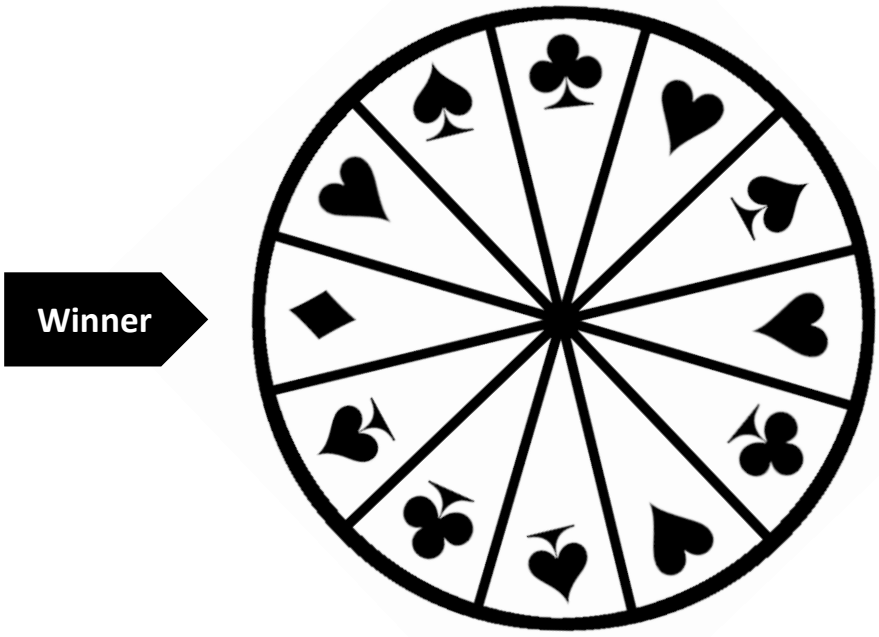
52% = 52 / 100 = (52 ÷ 4) / (100 ÷ 4) = 13 / 25 [B1]

$$\frac{13}{25}$$

(1 mark)

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Q6



If the wheel is spun again what is the chance of ◆ being the winner?
Give your answer as a fraction.

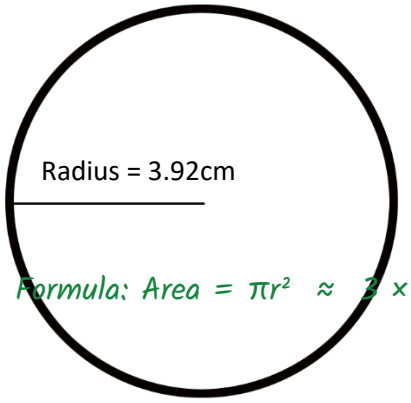
[BI]

Total sectors = 12, Diamond = 1 --> Probability = 1 / 12

$\frac{1}{12}$

(1 mark)

Q7



Formula: Area = πr^2 ≈ $3 \times 4^2 = 3 \times 16 = 48\ cm^2$ --> Option C

Which calculation gives an approximation of the area of this circle in cm^2?

(tick one box)

A 3 x 8 = ☐

B 3 x 4 = ☐

C 3 x 16 = ☑ [BI]

D 3 x 9 = ☐

(1 mark)

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Q8

Calculate $\frac{8-4^2}{8} =$

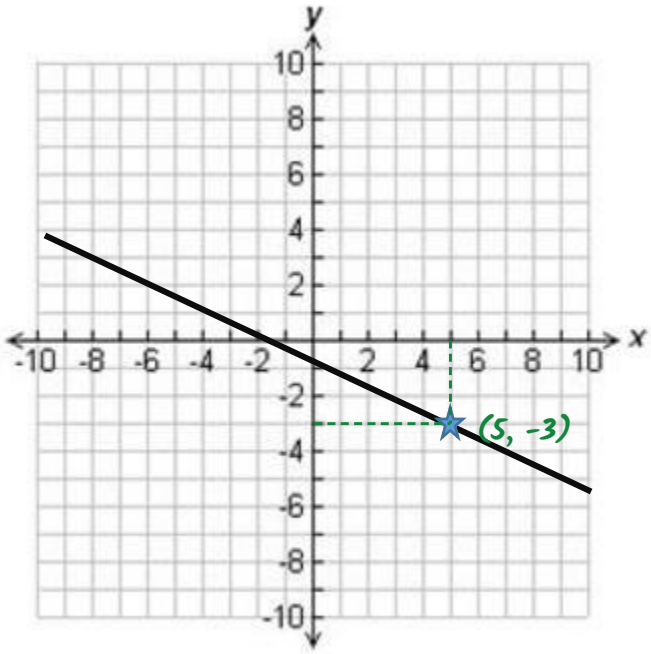
BIDMAS: $4^2 = 16$ --> $(8 - 16) \div 8 = -8 \div 8 = -1$

[B1]

-1

(1 mark)

Q9



What are the coordinates of the line at point ★?

(tick one box)

A (-5, 3) ☐

B (5, -3) ☑

C (-3, 5) ☐

D (3, -5) ☐

[B1]

(1 mark)

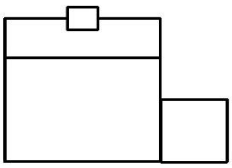
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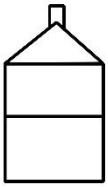
Q10

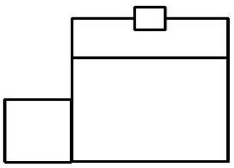


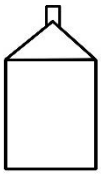
Which one of the following is the front elevation of the house from the view shown?

(tick one box)

A  ☐

B  ☑ [BI]

C  ☐

D  ☐

Front elevation shows the face with door and single upper window --> Option C

(1 mark)

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Q11

A commuter uses a bus and a train to get to work.

The train is more than 5 minutes late 1/6 of the times they use it

The bus is more than 5 minutes late 3/5 of the times they use it

What is the probability that **both** the bus and train will be more than 5 minutes late?

Show your working

P(train late) = 1/6, P(bus late) = 3/5

P(both late) = 1/6 × 3/5 = 3/30 = 1/10 (or 0.1)

[M1]

[A1]

1/10

(2 marks)

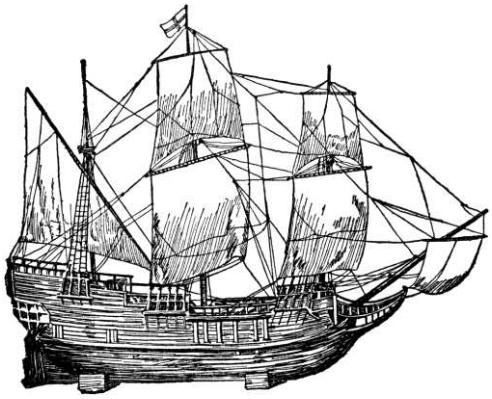
Q12

A model maker wants to make a model of a ship

He will use a scale of 1:50

The ship was 100ft long

1 foot = 0.3 m



How long will the model ship be in cm?

Show your working

Step 1: Real ship length = 100 ft × 0.3 m/ft = 30 m = 3000 cm

Step 2: Scale 1 : 50 --> Model length = 3000 cm ÷ 50 = 60 cm

(Check: 30 m ÷ 50 = 0.6 m = 60 cm)

[M1]

[M1]

[A1]

60 cm

(3 marks)

End of Section 1

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Level 2 Functional Skills Mathematics
Sample 3

City & Guilds
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Version 1-2

Duration: 1 hour 20 minutes
Total marks: 45 marks

MODEL SOLUTION SCORE
45 / 45 marks (100% Pass)
1 hr 20 mins · Verified by MFS

SECTION 2 – CALCULATOR PERMITTED

Candidate name (first, last)

First: MASTER

Last: FUNCTIONAL SKILLS

Candidate enrolment number: MFS4748

Date of birth (DDMMYYYY): 01012000

Assessment date (DDMMYYYY): 20102026

Centre number: MFS0001

Candidate signature and declaration*: Master Functional Skills

• If you have used any additional answer sheets write the number of additional sheets in this box. ☐
• Please ensure that you **staple** additional answer sheets to the **back** of this booklet, clearly labelling them with your full name, enrolment number, centre number and date in BLOCK CAPITALS.
• You must use a black or blue pen. You may use a pencil for charts and diagrams.
***I declare that I had no prior knowledge of the questions in this assessment and that I will not share information about the questions.**

Please check that your name is correctly printed on the candidate barcode label. If not, please tell the invigilator before the start of the exam.

You should have the following for this assessment

- a pen with black or blue ink
- a pencil
- an eraser
- a 30cm ruler
- a calculator



You must NOT use a protractor.

General instructions

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- You may use a dictionary.
- Show your working out (where required).
- Write all your working out and answers in this booklet.
- Check your calculations and check that your answers make sense.
- There are additional pages at the back of this booklet if you run out of space or ask the invigilator if you need additional sheets of paper.

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SECTION 2 – CALCULATOR PERMITTED

There are **45** marks in this section.

You should check all your work as you go along.

You may use a calculator.



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Q1

What is 12% of £5 300?

12% of £5300 = 0.12 × £5300 = £636

[B1]

£ 636

(1 mark)

Q2

What is 23.3 x 0.463 correct to three decimal places?

23.3 × 0.463 = 10.7879 --> to 3 d.p. = 10.788

[B1]

10.788

(1 mark)

Q3

Which one of the following lists is in decreasing order?

Decimals: 200/57 ≈ 3.51, 3 2/7 ≈ 3.29, 315/100 = 3.15, 1500/924 ≈ 1.62

(tick one box)

A	$\frac{315}{100}$	$\frac{1500}{924}$	$\frac{200}{57}$	$3\frac{2}{7}$	☐
B	$\frac{200}{57}$	$3\frac{2}{7}$	$\frac{315}{100}$	$\frac{1500}{924}$	☑
C	$\frac{200}{57}$	$\frac{1500}{924}$	$3\frac{2}{7}$	$\frac{315}{100}$	☐
D	$3\frac{2}{7}$	$\frac{1500}{924}$	$\frac{315}{100}$	$\frac{200}{57}$	☐

[B1]

3.51 > 3.29 > 3.15 > 1.62 (Decreasing)

(1 mark)

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Q4



What is the price of the phone before VAT of 20% is added?

Price with VAT = 120% = 1.20 × original price

Original price = £240 ÷ 1.20 = £200

[B1]

£200

(1 mark)

Q5

The results for a local election were

Candidate One	602	votes
Candidate Two	553	votes
Candidate Three	350	votes

What fraction of the votes did the winner get?

Give your answer in its simplest form.

Total votes = 602 + 553 + 350 = 1505 votes

Winner (Candidate 1) = 602 votes

Fraction = 602 / 1505 = (602 ÷ 301) / (1505 ÷ 301) = 2 / 5

[B1]

2/5

(1 mark)

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Q6

A parent fills in a medical form about her 2-year-old child.

The parent measures her child's height with a tape measure and writes that the child is 3m tall.

The receptionist at the surgery thinks that the parent has made a mistake.

Is the receptionist correct?

Is the receptionist correct? *(tick one box)* **Yes** ☑ **No** ☐

Explanation

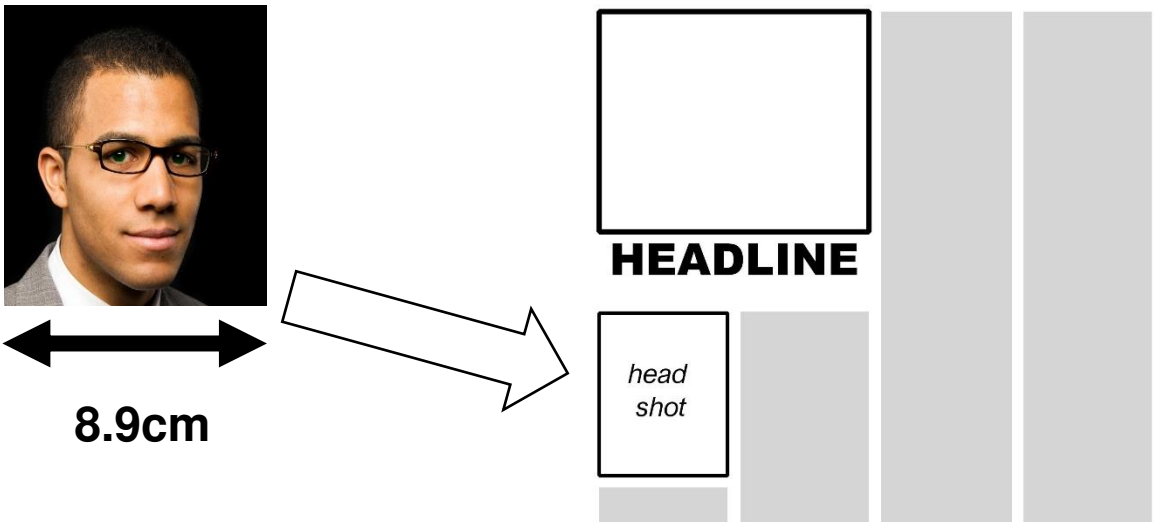
- 3m (300cm) is taller than a ceiling or standard doorframe (approx 2m). [A1]
- A typical 2-year-old child is around 85–90cm (less than 1 metre) tall.
- The parent likely misread feet for metres (3 feet ≈ 91cm is realistic).

(1 mark)

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Q7

An editor needs to change the width of this photo to fit in a magazine.



The photo needs to be 32 mm wide.

What percentage of the original width will the changed photo be?

Show your working

Step 1: Original width = 8.9 cm = 89 mm [M1]

Step 2: Percentage = (32 mm ÷ 89 mm) × 100 = 35.955...% ≈ 36.0% [A1]

36.0%

_____ %

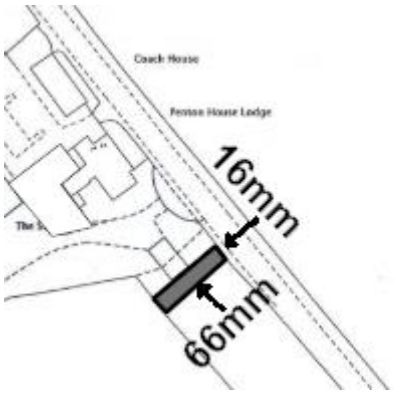
(2 marks)

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Q8

An estate agent needs to know the area of a plot of building land.

This plan shows the building land.



Scale 1:1250

= building land for sale

What is its area?

Show your working

Step 1: Plan dimensions are 16 mm by 66 mm

Real length = 66 mm × 1250 = 82,500 mm = 82.5 m

Real width = 16 mm × 1250 = 20,000 mm = 20.0 m

Step 2: Area = 82.5 m × 20.0 m = 1650 m²

[M1]

[M1]

1650

[A1] m²

(3 marks)

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Q9

An architect designs a shop refit.

The architect needs to work out the weight of an oak panel.

The oak panel measures 0.06 m x 0.9 m x 1.5 m

The density of oak is 700 kg per m^3

What is the weight of the oak panel in kg?

Show your working

Step 1: Volume = 1.5 m × 0.9 m × 0.06 m = 0.081 m^3 [M1]

Step 2: Weight = Volume × Density [M1]

Weight = 0.081 m^3 × 700 kg/m^3 = 56.7 kg [A1]

Weight of the oak panel 56.7 **kg**

(3 marks)

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Q10

A cook needs to cook a turkey to be ready by 6pm.

The cook will start cooking the turkey at 1:30pm

Turkey needs to be cooked for 45 minutes per kg of total weight plus 20 minutes and then left to rest for 45 minutes.

The cook has a 4.4kg turkey.

Has the cook allowed enough time? Explain your answer using figures.

Show your working

Has the cook allowed enough time? *(tick one box)* **Yes** ☑ **No** ☐

Explanation

- Cooking time = (4.4 kg × 45 mins) + 20 mins = 198 + 20 = 218 mins (3h 38m) [M1]
- Resting time = 45 mins
- Total time needed = 218 + 45 = 263 mins = 4 hours 23 minutes [M1]
- Ready time = 1:30 pm + 4h 23m = 5:53 pm [M1]
- Since 5:53 pm is before 6:00 pm (7 mins spare), the cook has allowed enough time. [A1]

(4 marks)

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Q11

A cafe makes its own coffee blend by mixing different coffee beans together.

It uses Brazilian beans and Kenyan beans in a ratio of 4:1

The manager needs to mix enough beans to make **120 cups of coffee.**

Each cup of coffee needs 85g of beans.

He needs to know what weights of beans to use.

What are the weights of the beans he needs?

Show your working

Step 1: Total beans = 120 cups × 85 g = 10,200 g = 10.2 kg [M1]

Step 2: Ratio 4 : 1 --> Total 5 parts. 1 part = 10.2 kg ÷ 5 = 2.04 kg [M1]

Step 3: Brazilian (4 parts) = 4 × 2.04 = 8.16 kg • Kenyan = 2.04 kg [M1]

Brazilian beans 8 [A1] **kg**

Kenyan beans 2.04 **kg**

(4 marks)

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Q12

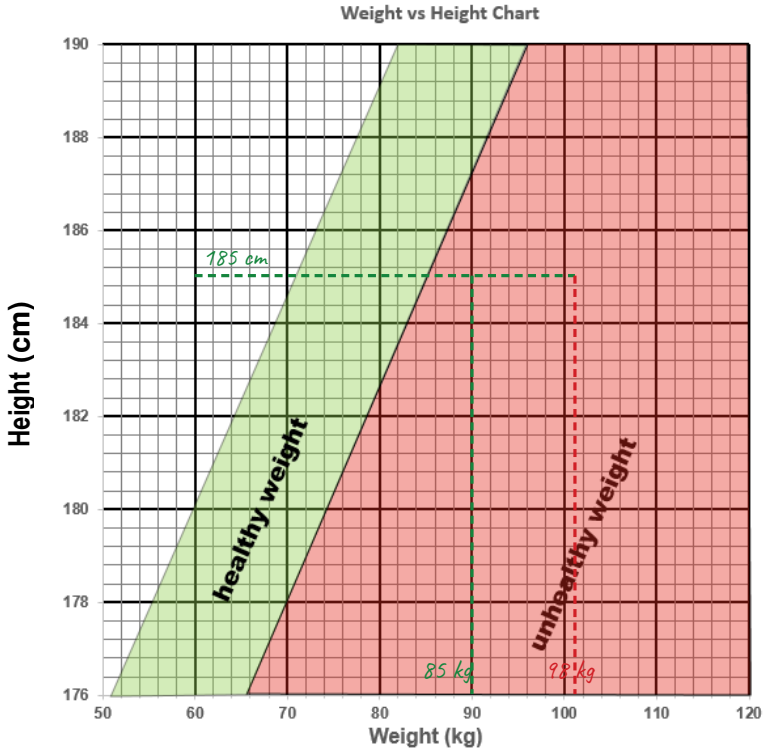
A man is 185cm tall and weighs 98kg. He wants to reach a healthy weight.

A doctor says he can lose 0.75kg a week if he follows a diet **and** exercises to burn 600 calories a week.

He plans his diet and decides to use Yoga exercise to burn 600 calories a week.

He will use one-hour Yoga classes that burn 200 calories. Each class costs £6.

He has £360 he can spend on Yoga classes



Does he have enough money to pay for the Yoga classes?

Explain your answer using figures.

Show your working

- Healthy weight = 85 kg --> Weight to lose = 98 - 85 = 13 kg [M1]
- Weeks: 13 ÷ 0.75 = 17.33 wks. Classes/wk: 600 ÷ 200 = 3 [M1]

Does he have enough money? *(tick one box)* **Yes** ☑ **No** ☐

Explanation

- Total classes = 17.33 × 3 = 52 classes (or 18 × 3 = 54 classes) [M1]
- Total cost = 52 × £6 = £312 (or 54 × £6 = £324) [M1]
- Since £312 < £360 (saving £48), he has enough money for the classes. [A1]

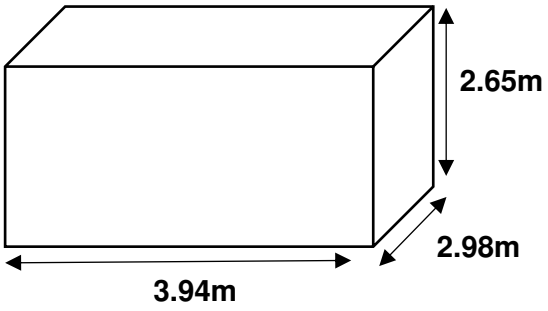
(5 marks)

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Q13

A house tenant has a budget of £75 to redecorate a room.

The room has two walls 2.98m long and 3.94m long and the wall is 2.65m high.



Sketch of room

Not to scale

She wants to paint one long wall and one short wall.

She wants to put wallpaper on the other short and long walls.

A 2.5L tin of paint costs £16 and covers 30m^2

A roll of wallpaper is 52cm wide and 10 m long.



Wallpaper

£10 per roll

Can the tenant decorate the room for the budget set?

Explain your answer using figures.

Can the tenant decorate the room for the budget set? *(tick one box)* **Yes** ☑ **No** ☐

Show your working

- Painted area = (3.94 + 2.98) × 2.65 = 18.338 m² --> 1 tin = £16 [M1]
- Wallpaper area = 18.338 m² · 1 roll = 0.52 × 10 = 5.2 m² [M1]
- Rolls needed = 18.338 ÷ 5.2 = 3.53 --> 4 rolls × £10 = £40 [M1]

Explanation

- Total cost = £16 (paint) + £40 (wallpaper) = £56 [M1]
- Budget is £75. Since £56 < £75 (under by £19), tenant can decorate. [A1]

(6 marks)

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Q14

A vet nurse in a pet rescue centre records weights and ages of kittens brought to the centre.

Weights and ages of kittens brought to the centre				
weight (g)	**age (days)**		**weight (g)**	**age (days)**
600	50		660	50
340	28		720	56
480	42		420	28
360	20		180	14

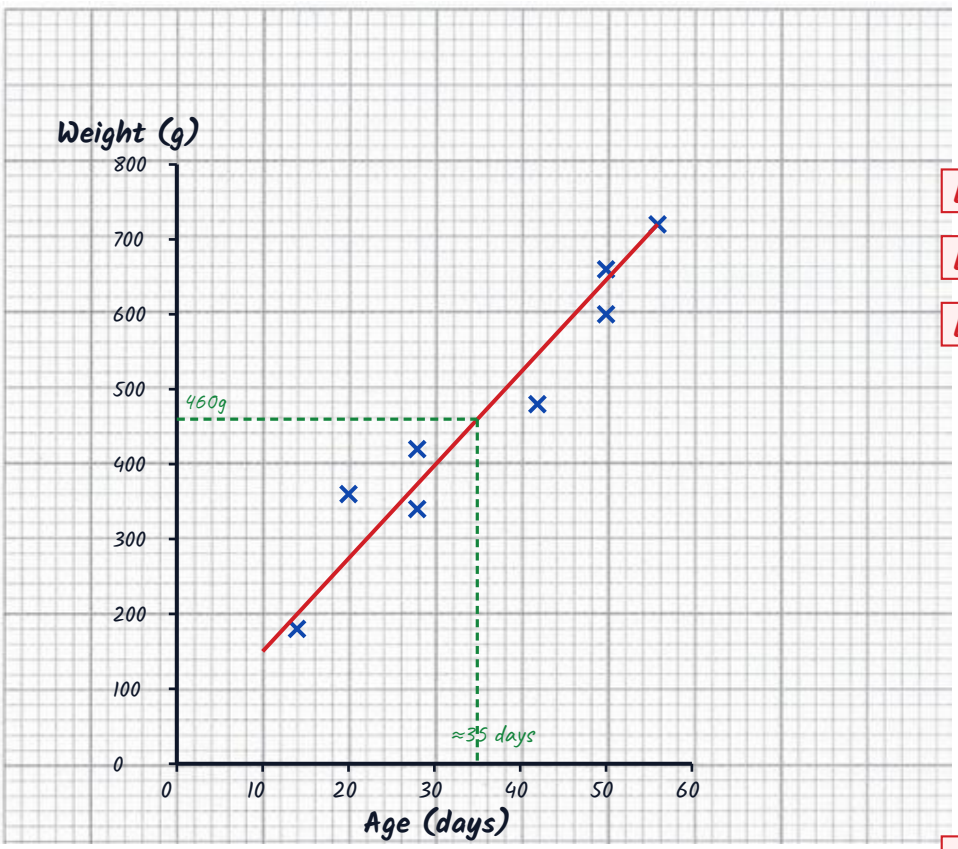
One day the centre receives an abandoned kitten. It weighs 460g

The nurse needs to estimate the age of the kitten.

Draw a suitable graph and trend line.

Estimate the age of the abandoned kitten.

Show on the graph how you made the estimation.



[M1]

[M1]

[A1]

[M1]

35 days ÷ 7 days/week = 5 weeks

[A1]

Estimated age of rescued kitten 5 **weeks**

(5 marks)

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Q15

A city council wants to encourage car sharing to cut pollution.

It takes a survey of cars entering the city and find these results.

Average people in a car entering the city	
Mean	**Mode**
1.65	1

It introduces a congestion charge for cars with only one person in the car.

It wants to know if the congestion charge is effective in making people share cars.

It takes another survey of the number of people in cars entering the city.

Number of people in a car	Number of cars **after** the congestion charge is introduced (frequency)	Cars × People (fx)
1	17	1 × 17 = 17
2	54	2 × 54 = 108
3	14	3 × 14 = 42
4	8	4 × 8 = 32
5	2	5 × 2 = 10
6+	0	6+ × 0 = 0
	Total cars = 95	Total fx = 209

Is the congestion charge effective?

Make two comments using the figures provided

Is the congestion charge effective? *(tick one box)* **Yes** ☑ **No** ☐

Show your working Mean = 209 ÷ 95 = 2.20 Mode = 2 (frequency 54) [M1]

Comment 1

Yes, the congestion charge is effective because the mean number of people per car increased from 1.65 to 2.20 (2.20 > 1.65). [M1] [A1]

Comment 2

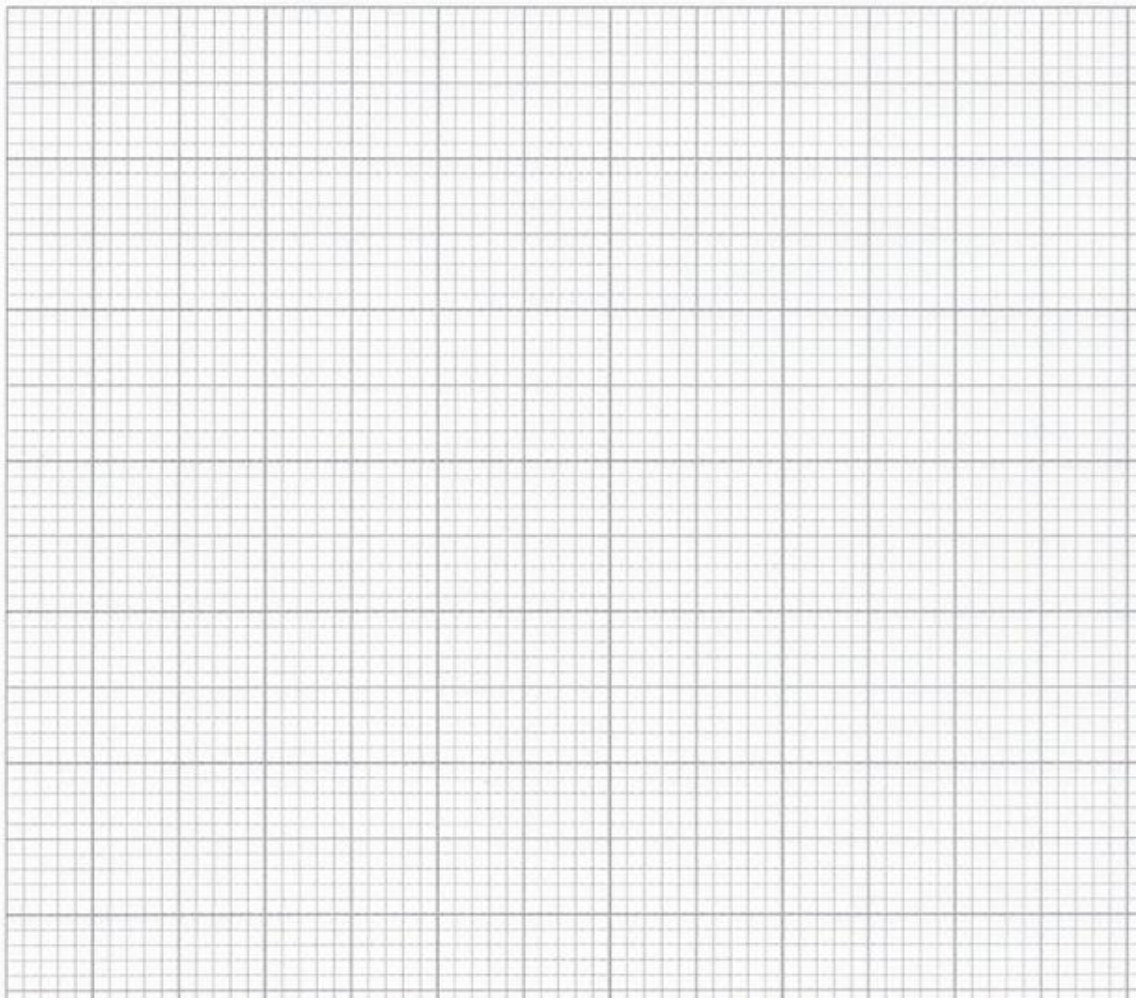
[B1]

The modal number of people in a car increased from 1 to 2 (2 > 1), showing that more people are now car sharing. [A1] [A1]

(6 marks)

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Spare graph paper for Question 14



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Extra space for working out and answers

End of section 2

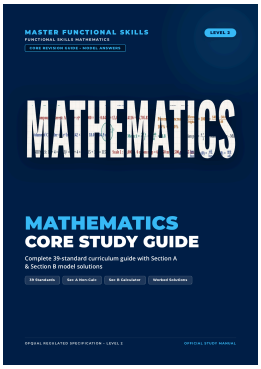
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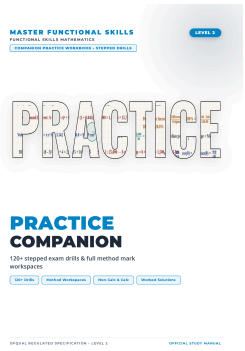


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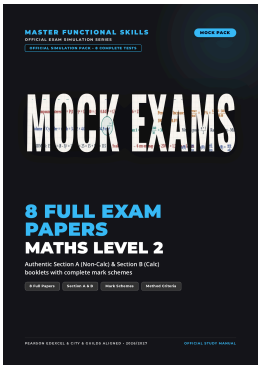


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